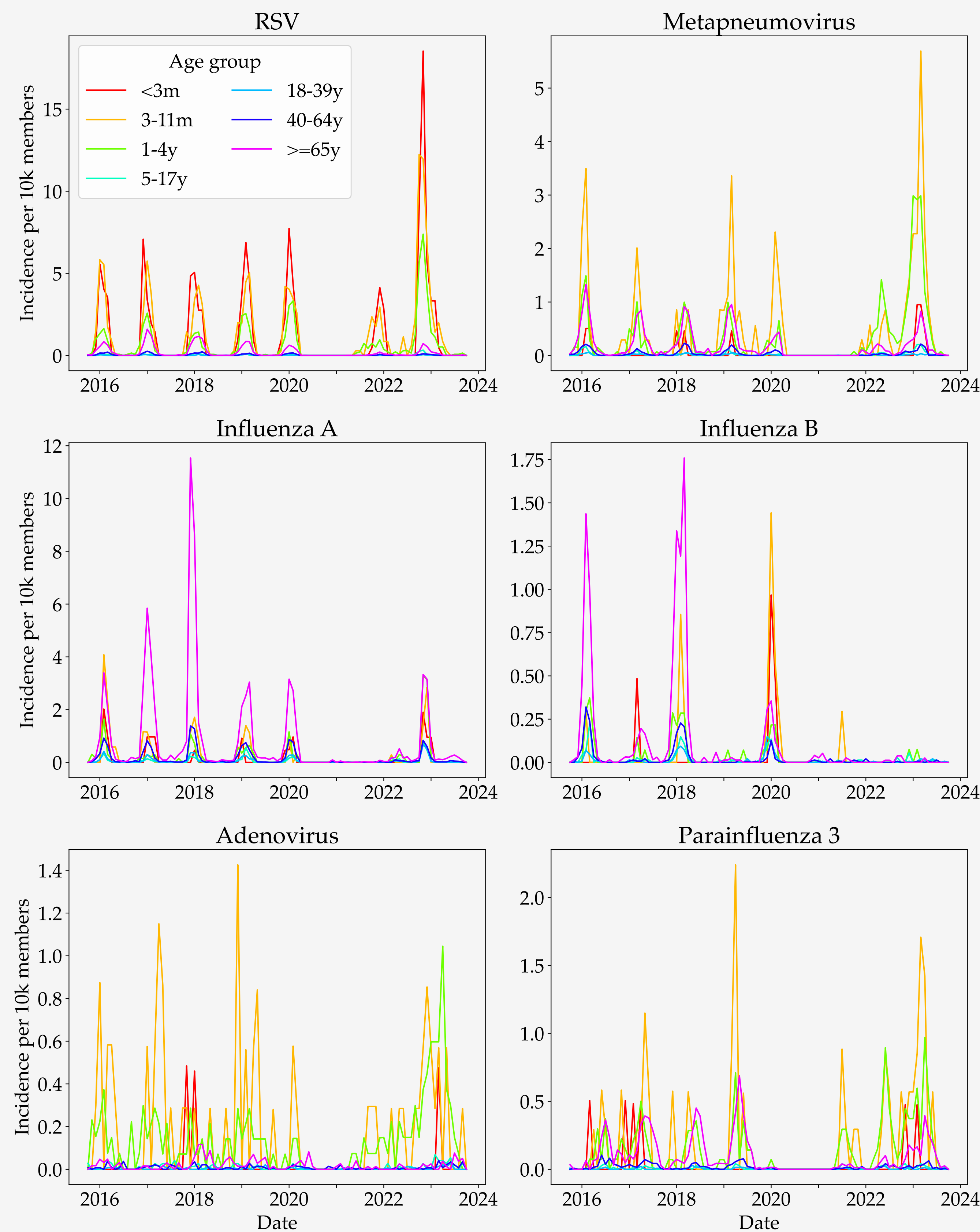
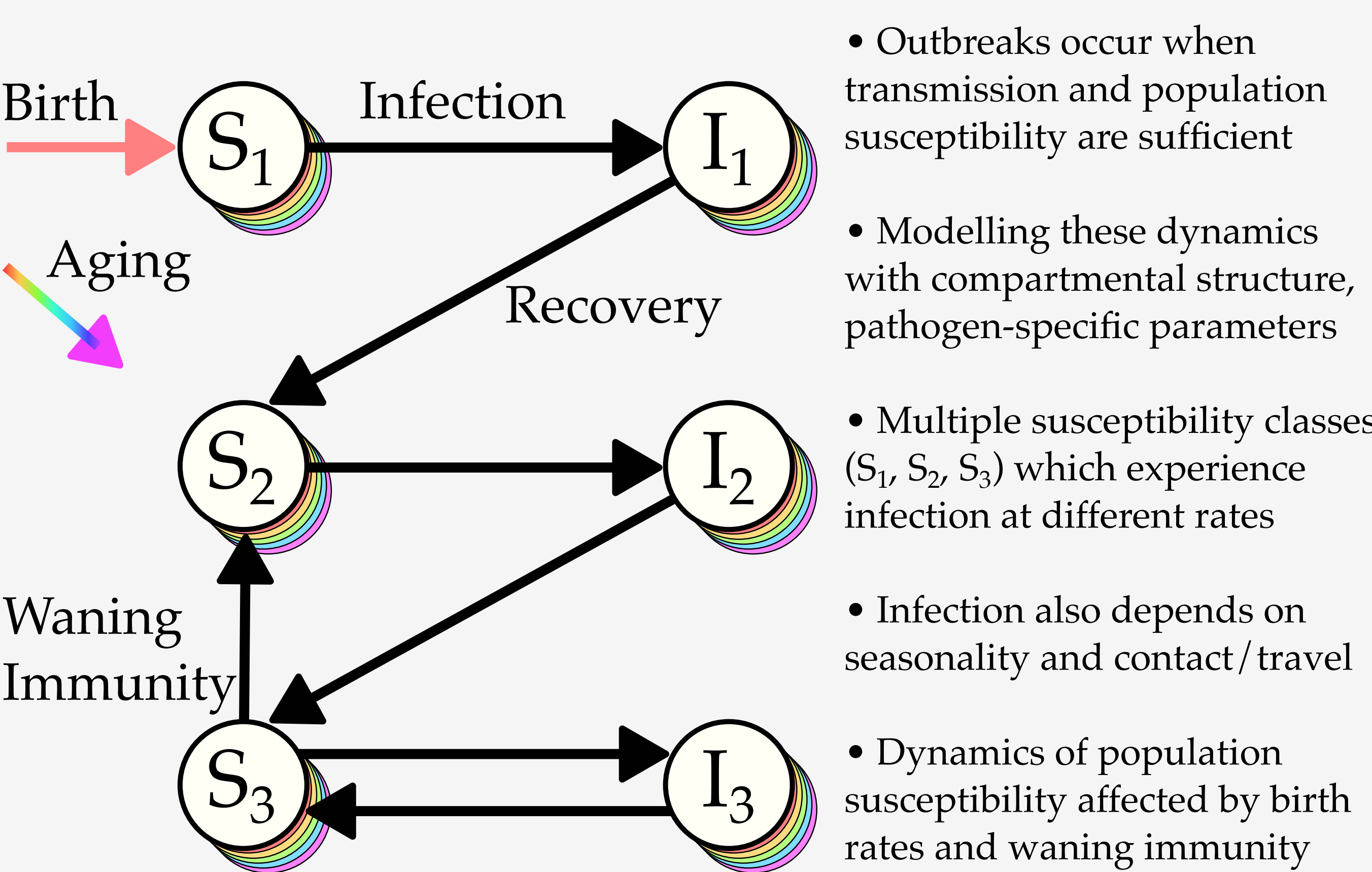


Pathogens rebounded differently

- Kaiser Permanente Southern California data allows us to track respiratory disease incidence using test-confirmed hospitalized cases
- After supression during the pandemic, each pathogen re-emerged differently



Outbreaks have dynamic causes



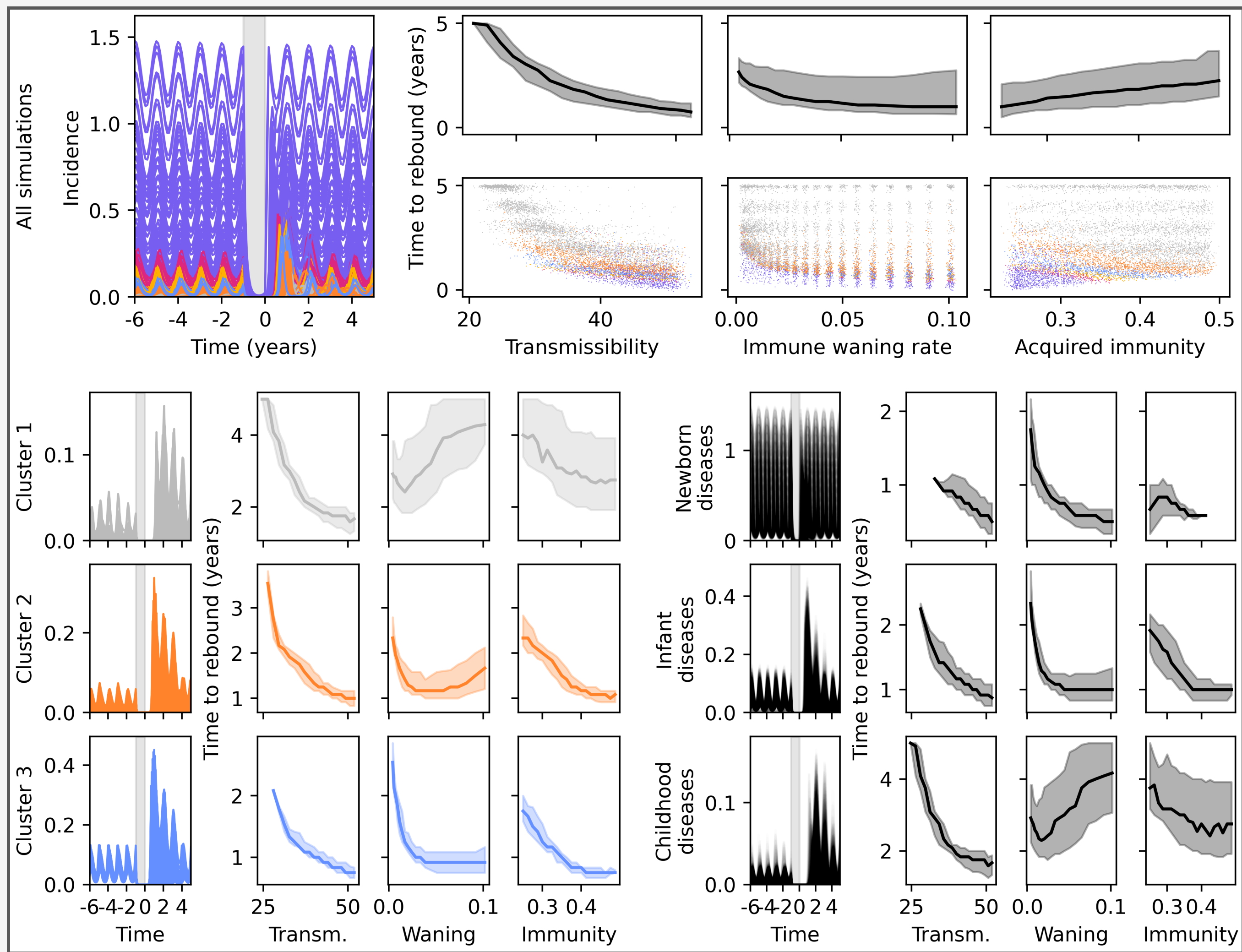
- Outbreaks occur when transmission and population susceptibility are sufficient
- Modelling these dynamics with compartmental structure, pathogen-specific parameters
- Multiple susceptibility classes (S_1, S_2, S_3) which experience infection at different rates
- Infection also depends on seasonality and contact/travel
- Dynamics of population susceptibility affected by birth rates and waning immunity

A mechanistic model of the post-COVID-19 pandemic re-emergence of respiratory viruses in Southern California

Benjamin J Singer¹, Vennis Hong², Joseph Lewnard¹, Sara Tartof²

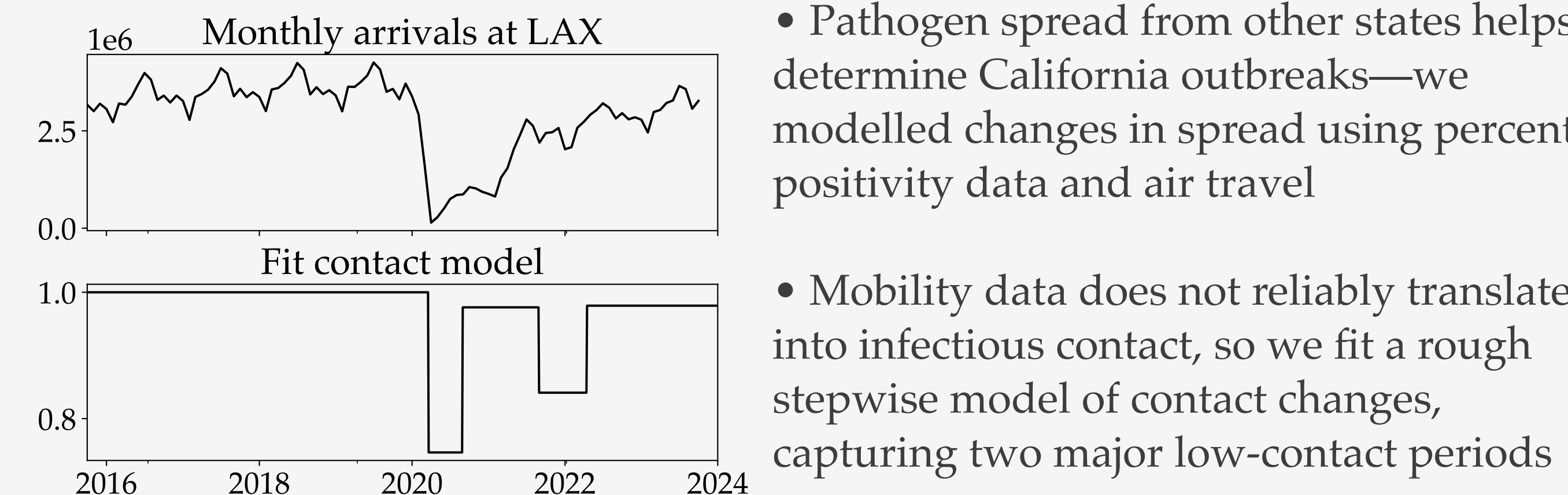
¹ Division of Infectious Diseases, School of Public Health, University of California, Berkeley
² Department of Research and Evaluation, Kaiser Permanente Southern California

Parameters have non-linear effects



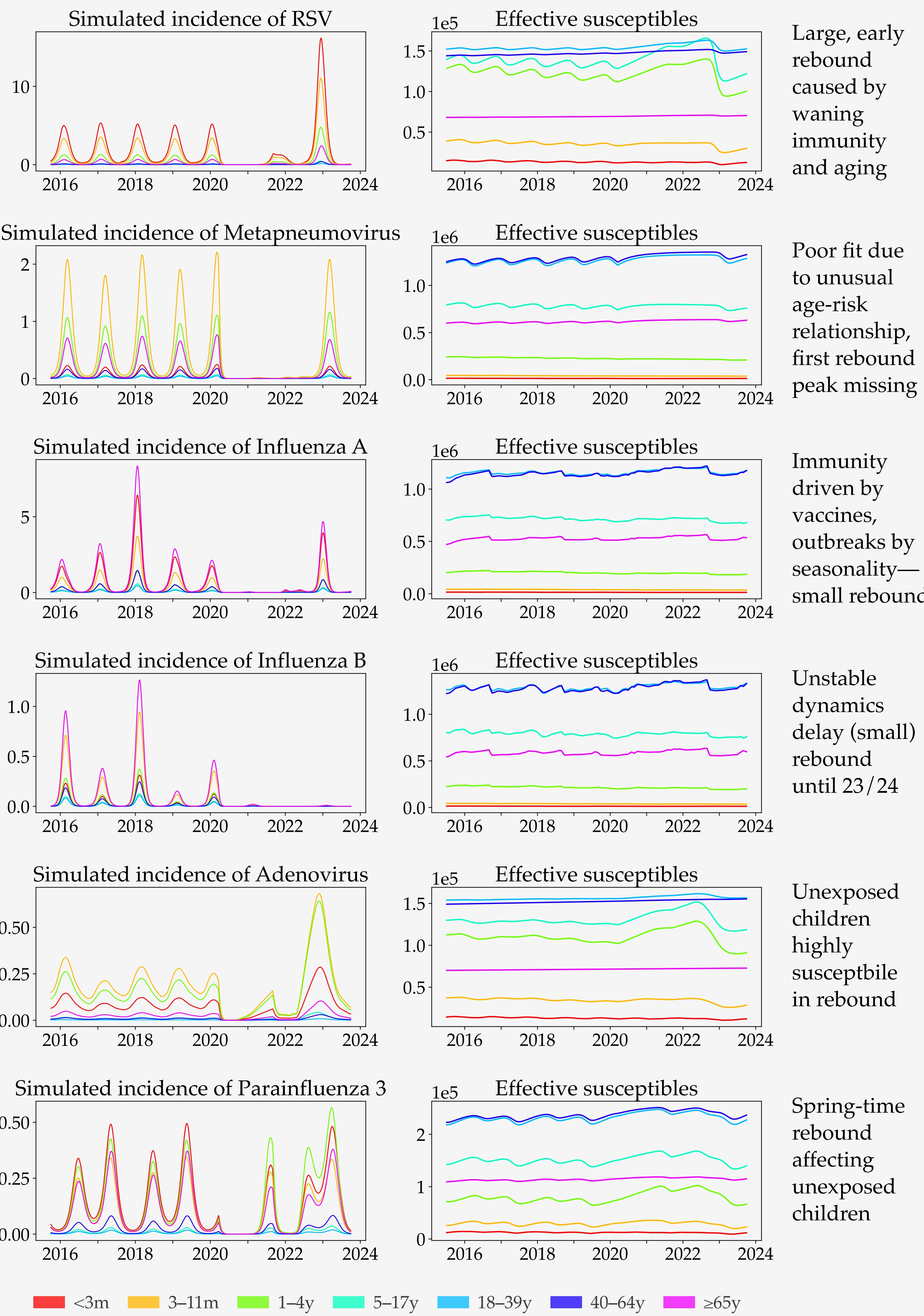
- Multiple parameters can change the time for a simulated disease to rebound after a period of suppression
- Trajectories and points are colored according to a clustering algorithm based on pre-suppression dynamics, showing how similar diseases can rebound differently
- Acquired immunity has a different effect on time to rebound overall vs within clusters, showing the complex role of different parameters on re-emergence

COVID-19 reduced contact and travel



- Pathogen spread from other states helps determine California outbreaks—we modelled changes in spread using percent positivity data and air travel
- Mobility data does not reliably translate into infectious contact, so we fit a rough stepwise model of contact changes, capturing two major low-contact periods

Model fits reveal different dynamics



- Large, early rebound caused by waning immunity and aging
- Poor fit due to unusual age-risk relationship, first rebound peak missing
- Immunity driven by vaccines, outbreaks by seasonality—small rebound
- Unstable dynamics delay (small) rebound until 23/24
- Unexposed children highly susceptible in rebound
- Spring-time rebound affecting unexposed children

Next steps for public health impact

- Models can explain different re-emergence characteristics depending on population dynamics, waning immunity, and seasonality
- These differing parameters correspond to different population dynamics of age-dependent susceptibility, which drive outbreaks
- By quantifying uncertainty in the fits we will show what information on disease spread has been gained by measuring rebounds
- In future these models could be deployed to understand which age groups drive disease spread and anticipate disease rebounds after periods of suppression